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Q3.2023

Crypto Market Quarterly Review

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Executive Summary: The Calm Before the Surge

The best word to describe crypto during the third quarter of 2023 is "quiet."

Prices drifted downwards, with the Bitwise 10 Large Cap Crypto Index falling roughly 13%. Trading volumes were the lowest in nearly three years. Even big news stories—usually a constant in crypto—were few and far between.

Some see this quiet period as a sign of malaise, but after looking through the numbers I think that's wrong. The data in this quarter's report, which features more than 50 charts and key metrics on the forces moving the crypto markets, shows an industry holding steady, but in a much stronger position than in previous cycles. It shows crypto is a coiled spring.

For example:

- **Crypto prices**, while down for the quarter, remain sharply up for the year: The Bitwise 10 Large Cap Crypto Index closed Q3 up 50% on a year-to-date basis, making crypto the best-performing major asset class in the world this year.
- **Blockchain revenue,** the best measure of crypto's real-world utility, is up significantly from the last cycle. In 2019, the last time crypto was emerging from a bear market, Ethereum did $35 million in revenue. So far this year it's done $1.75 billion.
- **Interest in decentralized finance (DeFi)**—whether measured by exchange trading volume, lending volume, stablecoin AUM, or stablecoin activity—is down from peak levels but still remarkably strong. For example, stablecoins settled nearly $1.5 trillion in transactions in Q3, which is more volume than all of the credit card transactions that Mastercard typically processes each quarter.[1]
- **Crypto equities** are among the fastest-growing stocks in the market based on revenue growth. In 2023, the median public crypto company is projected to post more than double the revenue growth of any sector in the S&P 500.

Looking ahead, we see a number of pending catalysts that could catapult us into a strong bull market, from the possible approval of a spot bitcoin ETF later this year (yes, in 2023) to the bitcoin halving coming up in April 2024.

We hope you enjoy the report.



Matt Hougan
Chief Investment Officer
Bitwise Asset Management

(1) See Mastercard's quarterly conference call presentations, available at https://investor.mastercard.com/events-and-presentations/presentations/default.aspx.

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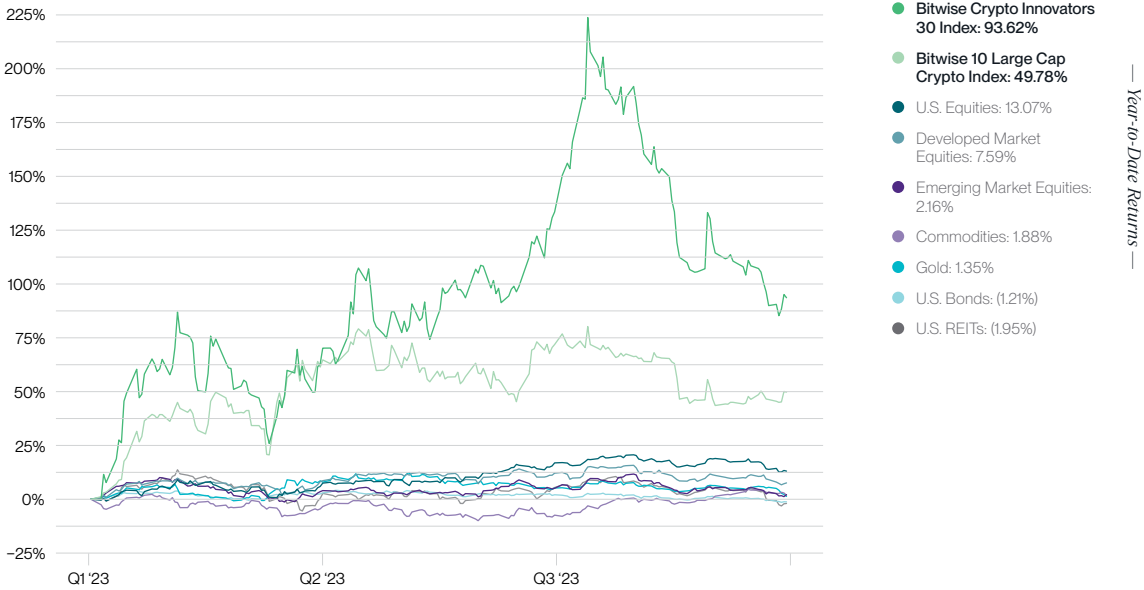
Howie Levine
Sr. Regional Consultant

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Market Performance: Crypto vs. Major Asset Classes



Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

Note: Asset classes are represented by the following. U.S. Equities: S&P 500 Total Return Index. DM Equities: MSCI EAFE Gross Total Return USD Index. EM Equities: MSCI Emerging Markets Gross Total Return USD Index. Commodities: Deutsche Bank DBIQ Optimum Yield Diversified Commodity Index Total Return. Gold: Gold spot price. U.S. Bonds: Bloomberg U.S. Aggregate Bond Index. U.S. REITs: MSCI U.S. REIT Gross Total Return Index.

All calculations are total return, including dividends for the stated period. Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Index performance does not include the fees and expenses that are charged by any fund. Fund returns may differ materially from the returns of an index. Past performance is no guarantee of future results. Please refer to additional important disclosures at the back of this document. For definitions of the indexes and assets listed here, see p.43.

Market Performance: Crypto vs. Major Asset Classes

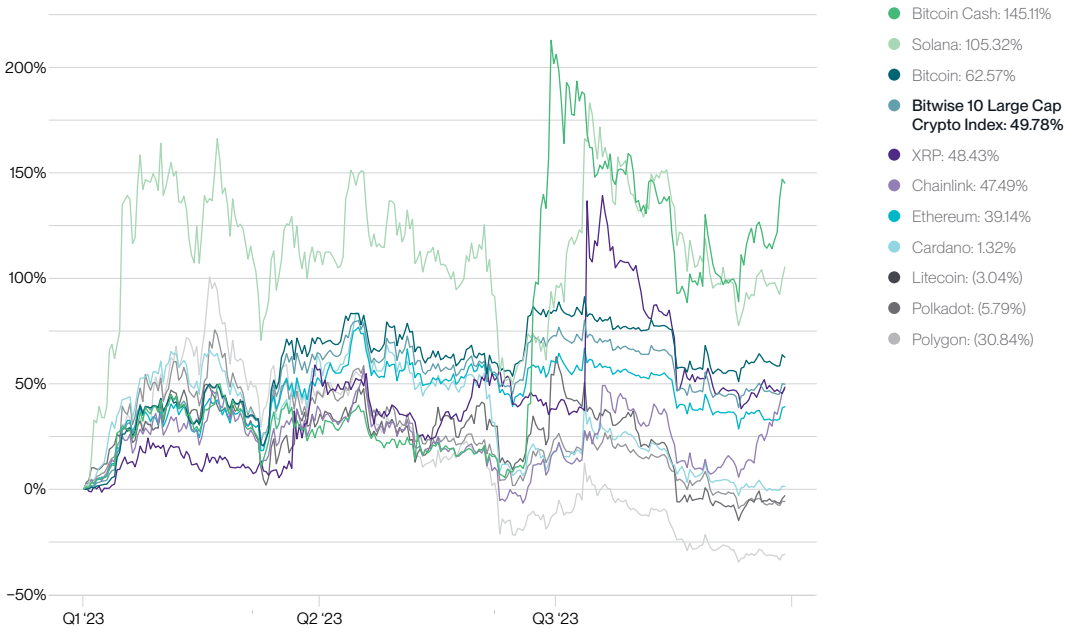
	— Cumulative —					— Annualized —	
	Q3 2023	YTD	1 YR	3 YR	5 YR	3 YR	5YR
Bitwise Crypto Innovators 30 Index (XBITQG)	(17.07%)	**93.62%**	3.20%	—	—	—	—
Bitwise 10 Large Cap Crypto Index (XBITW)	(12.87%)	**49.78%**	26.47%	144.92%	217.56%	34.83%	25.97%
U.S. Equities (SPXT)	(3.27%)	**13.07%**	19.79%	33.65%	60.44%	10.16%	9.90%
Developed Market Equities (M2EA)	(4.05%)	**7.59%**	26.31%	20.04%	20.17%	9.14%	4.73%
Emerging Market Equities (M2EF)	(2.79%)	**2.16%**	12.17%	(3.97%)	4.77%	2.62%	1.27%
Commodities (DBLCDBCT)	9.99%	**1.88%**	5.90%	97.70%	50.28%	23.28%	6.56%
Gold (XAU)	(3.68%)	**1.35%**	11.32%	(1.97%)	55.23%	(0.64%)	8.89%
U.S. Bonds (LBUSTRUU)	(3.23%)	**(1.21%)**	0.64%	(14.82%)	0.51%	(3.99%)	0.77%
U.S. REITs (RMSG)	(7.02%)	**(1.95%)**	4.54%	18.09%	14.89%	5.70%	2.81%

Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

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Performance of the Bitwise 10 Large Cap Crypto Index and Its Constituents



Source: Bitwise Asset Management with data from CoinGecko. Data as of September 29, 2023.

The Bitwise 10 Large Cap Crypto Index captures the 10 largest eligible crypto assets by free-float-adjusted market capitalization, screened and monitored for certain risks. Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Index performance and crypto asset performance do not include the fees and expenses that are charged by any fund. Fund returns may differ materially from the returns of an index or crypto asset. Past performance is no guarantee of future results. Please refer to additional important disclosures at the back of this document.

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Performance of the Bitwise 10 Large Cap Crypto Index and Its Constituents

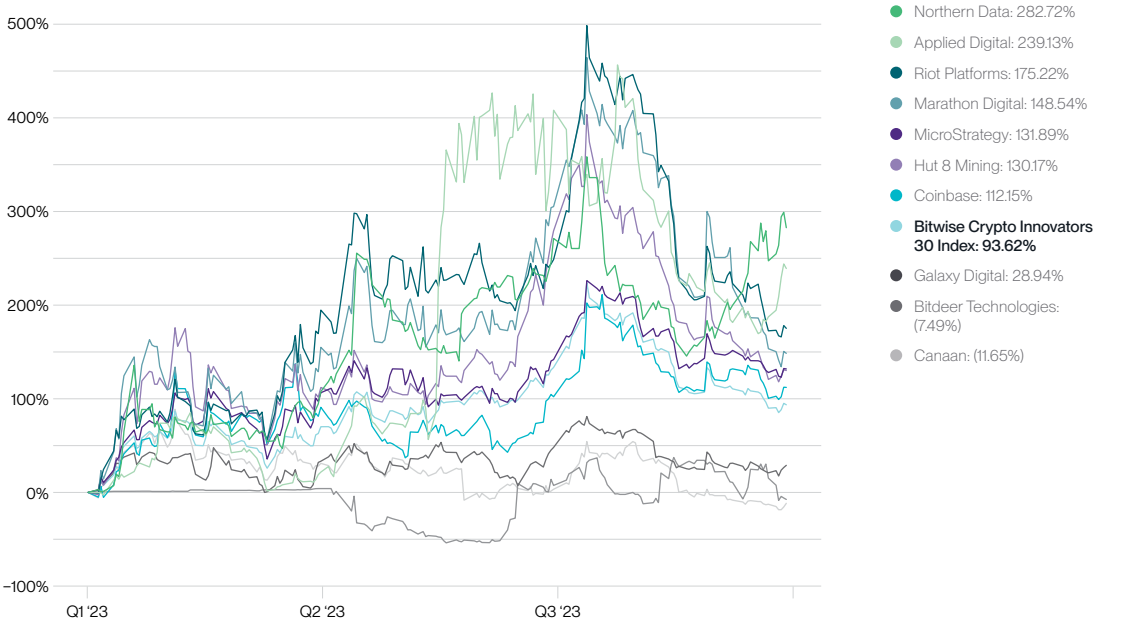
		— Cumulative —					— Annualized —	
	Index Weight	Q3 2023	YTD	1 YR	3 YR	5 YR	3 YR	5YR
Bitwise 10 Large Cap Crypto Index (XBITW)	—	(12.87%)	49.78%	26.06%	144.92%	217.56%	34.83%	25.97%
Bitcoin (BTC)	**65.99%**	(11.33%)	62.57%	38.17%	151.23%	303.22%	35.98%	32.12%
Ethereum (ETH)	**25.29%**	(13.54%)	39.14%	24.45%	368.22%	649.02%	67.37%	49.52%
XRP (XRP)	**3.46%**	11.66%	48.43%	9.67%	122.46%	(1.25%)	34.57%	(0.27%)
Cardano (ADA)	**1.11%**	(12.86%)	1.32%	(42.31%)	150.21%	—	35.61%	—
Solana (SOL)	**1.11%**	6.02%	105.32%	(40.01%)	611.84%	—	92.14%	—
Polkadot (DOT)	**0.66%**	(20.39%)	(5.79%)	(34.65%)	(5.42%)	—	(1.83%)	—
Polygon (MATIC)	**0.62%**	(20.23%)	(30.84%)	(32.57%)	2,556.97%	—	198.70%	—
Litecoin (LTC)	**0.61%**	(37.25%)	(3.04%)	22.76%	44.02%	6.97%	12.85%	1.34%
Bitcoin Cash (BCH)	**0.58%**	(21.64%)	145.11%	93.05%	4.78%	(56.50%)	1.57%	(15.32%)
Chainlink (LINK)	**0.57%**	27.53%	47.49%	6.92%	(17.75%)	2,297.55%	(6.31%)	88.65%

Source: Bitwise Asset Management with data from CoinGecko. Data as of September 29, 2023.

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Performance of the Bitwise Crypto Innovators 30 Index and Its Top 10 Constituents



Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

Note: Bitdeer Technologies (BTDR) went public via SPAC. The SPAC company was trading before the merger closed and BTDR went public on April 14, 2023.

The Bitwise Crypto Innovators 30 Total Return Index is an equity index that provides focused exposure to companies that are building the future of the crypto-asset-enabled decentralized economy. Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Index performance and equity performance do not include the fees and expenses that are charged by any fund. Fund returns may differ materially from the returns of an index or individual equity. Past performance is no guarantee of future results. Please refer to additional important disclosures at the back of this document.

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Performance of the Bitwise Crypto Innovators 30 Index and Its Top 10 Constituents

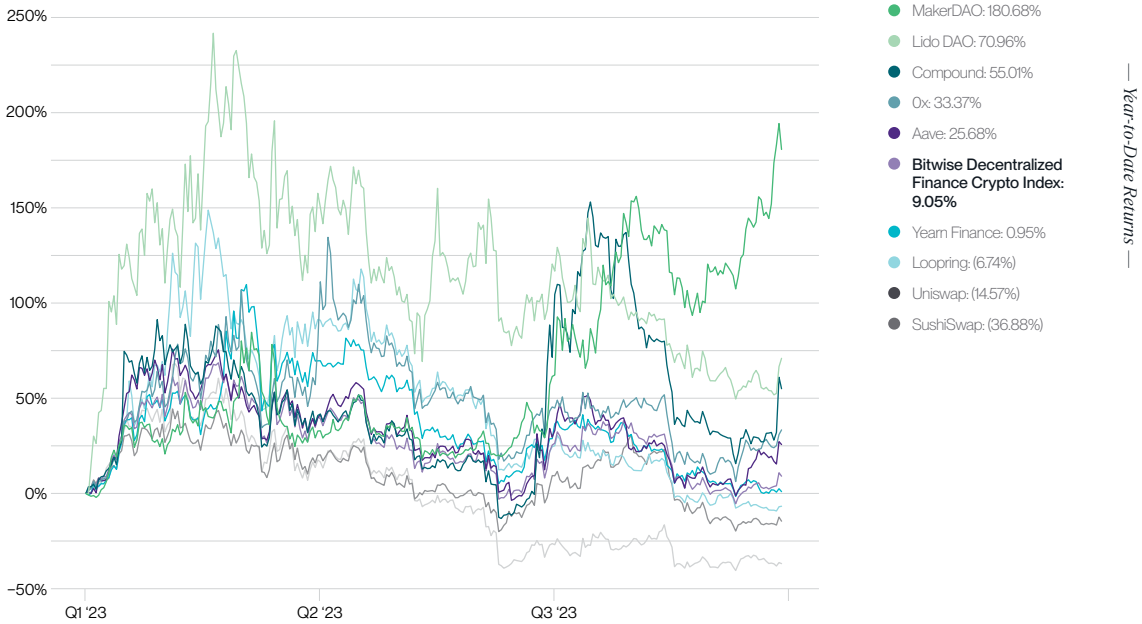
		— Cumulative —					— Annualized —	
	Index Weight	Q3 2023	YTD	1 YR	3 YR	5 YR	3 YR	5YR
Bitwise Crypto Innovators 30 Index (XBITQG)	—	(17.07%)	93.62%	3.20%	—	—	—	—
Coinbase (COIN)	**10.68%**	4.93%	112.15%	21.21%	—	—	—	—
MicroStrategy (MSTR)	**10.36%**	(4.13%)	131.89%	54.98%	118.04%	133.45%	29.70%	18.46%
Marathon Digital (MARA)	**7.86%**	(38.67%)	148.54%	(21.08%)	333.67%	191.30%	63.15%	23.81%
Northern Data (NB2 GR)	**7.01%**	2.66%	282.72%	118.19%	(56.32%)	(24.15%)	32.29%	5.75%
Applied Digital (APLD)	**5.39%**	(33.26%)	239.13%	290.00%	—	—	—	—
Galaxy Digital (GLXY CN)	**4.93%**	(12.91%)	28.94%	(14.41%)	48.07%	109.66%	13.99%	15.94%
Canaan (CAN)	**4.57%**	(14.55%)	(11.65%)	(44.17%)	(2.67%)	—	(0.90%)	—
Riot Platforms (RIOT)	**4.22%**	(21.07%)	175.22%	31.59%	245.56%	155.62%	51.24%	20.62%
Hut 8 Mining (HUT CN)	**4.17%**	(39.32%)	130.17%	8.10%	175.26%	(8.87%)	40.19%	(1.84%)
Bitdeer Technologies (BTDR)	**3.79%**	(13.93%)	(7.49%)	(6.13%)	—	—	—	—

Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

The Bitwise Crypto Innovators 30 Total Return Index is an equity index that provides focused exposure to companies that are building the future of the crypto-asset-enabled decentralized economy. Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Index performance and equity performance do not include the fees and expenses that are charged by any fund. Fund returns may differ materially from the returns of an index or individual equity. Past performance is no guarantee of future results. Please refer to additional important disclosures at the back of this document.

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Performance of the Bitwise Decentralized Finance Crypto Index and Its Constituents



Source: Bitwise Asset Management with data from CoinGecko. Data as of September 29, 2023.

The Bitwise Decentralized Finance Crypto Index is designed to provide investors with a clear, rules-based, and transparent way to track the value of the rapidly emerging Decentralized Finance space. Performance of an index is not illustrative of any particular investment. It is not possible to invest directly in an index. Index performance and crypto asset performance do not include the fees and expenses that are charged by any fund. Fund returns may differ materially from the returns of an index or crypto asset. Past performance is no guarantee of future results. Please refer to additional important disclosures at the back of this document.

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Performance of the Bitwise Decentralized Finance Crypto Index and Its Constituents

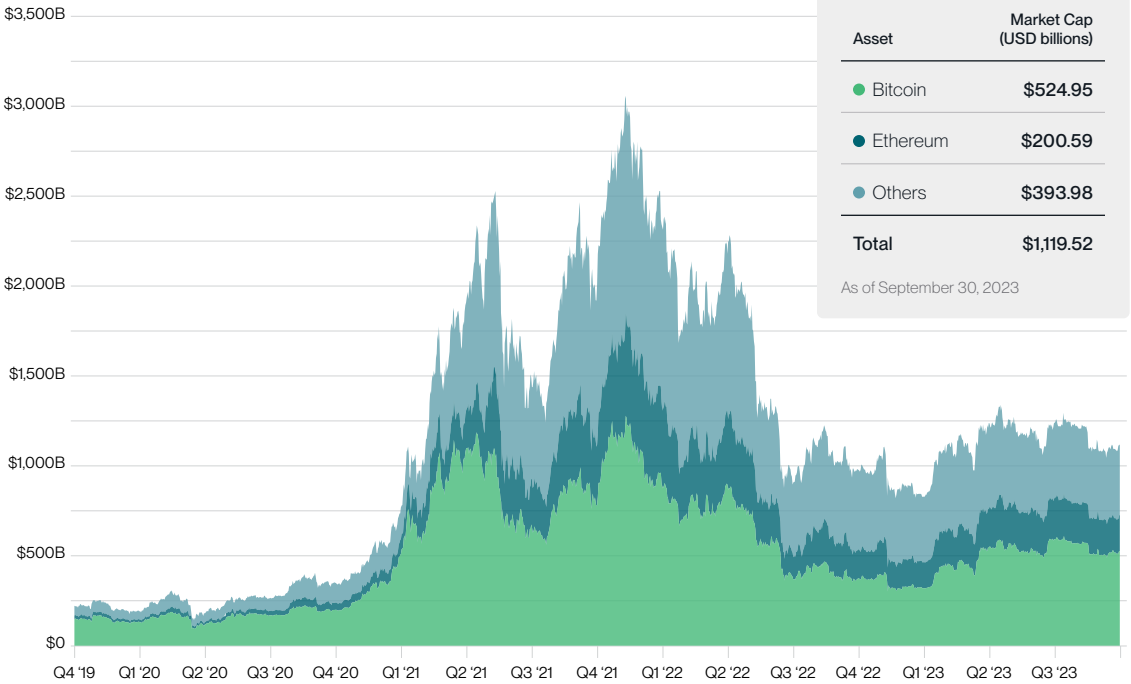
		— Cumulative —					— Annualized —	
	Index Weight	Q3 2023	YTD	1 YR	3 YR	5 YR	3 YR	5YR
Bitwise Decentralized Finance Crypto Index (XBITDEFI)	—	(7.32%)	9.05%	(18.56%)	—	—	—	—
Uniswap (UNI)	**40.32%**	(15.05%)	(14.57%)	(29.80%)	9.14%	—	1.87%	—
Lido DAO (LDO)	**18.42%**	(13.60%)	70.96%	10.06%	—	—	—	—
MakerDAO (MKR)	**16.63%**	88.85%	180.68%	105.68%	182.98%	—	38.66%	—
Aave (AAVE)	**11.82%**	(3.45%)	25.68%	(8.80%)	—	—	—	—
Compound (COMP)	**4.44%**	(17.79%)	55.01%	(22.45%)	(64.33%)	—	(29.09%)	—
Loopring (LRC)	**2.68%**	(22.01%)	(6.74%)	(39.38%)	(16.17%)	74.37%	(6.28%)	11.23%
Yearn Finance (YFI)	**2.08%**	(20.76%)	0.95%	(35.42%)	(77.77%)	—	(39.54%)	—
0x (ZRX)	**1.94%**	(9.20%)	33.37%	(28.35%)	(52.39%)	(71.21%)	(21.17%)	(21.47%)
SushiSwap (SUSHI)	**1.67%**	(11.12%)	(36.88%)	(44.88%)	(52.88%)	—	(22.83%)	—

Source: Bitwise Asset Management with data from CoinGecko. Data as of September 29, 2023.

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Total Crypto Market Capitalization



Asset	Market Cap (USD billions)
Bitcoin	$524.95
Ethereum	$200.59
Others	$393.98
Total	**$1,119.52**

As of September 30, 2023

Source: Bitwise Asset Management with data from The Block and CoinGecko. Data from Q4 2019 to Q3 2023.
Note: "Others" includes all crypto assets for which data is available from CoinGecko and The Block, including stablecoins, excluding Bitcoin and Ethereum.

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Bitcoin: Market Performance and Four-Year Cycles

"The Bitcoin Cycle"		*"The Ethereum Cycle"*		*"The Early Application Cycle"*		*"The Mainstream Cycle"*	
Year	BTC	Year	BTC	Year	BTC	Year	BTC
2011	1,319%	2015	34%	2019	94%	2023 (YTD)	63%
2012	217%	2016	123%	2020	308%	*2024*	*?*
2013	5,537%	2017	1,414%	2021	57%	*2025*	*?*
2014	(58%)	2018	(75%)	2022	(64%)	*2026*	*?*

Source: Bitwise Asset Management. Data from December 31, 2010 to September 29, 2023

Note: The names given to the four-year cycles represent our assessment of the forces that most contributed to Bitcoin's performance during that period.

Performance information is provided for informational purposes only. Returns reflect the return of Bitcoin itself, and not of any fund or account, and do not include any fees. Backward-looking performance cannot predict how any investment strategy will perform in the future. Future crypto cycles may not be four years long; the four-year increment is based on historical data for illustrative purposes and is not a prediction of future results. This material represents an assessment of the market environment at a specific time and is not intended to be a forecast of future events, or a guarantee of future results.

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Performance of Major Assets and Asset Classes

2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023 (YTD)
Bitcoin 1,319.09%	Bitcoin 217.26%	Bitcoin 5,537.40%	U.S. REITs 30.38%	Bitcoin 33.74%	Bitcoin 123.31%	Bitcoin 1,414.10%	U.S. Bonds 0.01%	Bitcoin 93.95%	Bitcoin 308.17%	Bitcoin 57.25%	Commodity 21.23%	Bitcoin 62.57%
Gold 10.10%	EM Equity 18.63%	U.S. Equity 32.39%	U.S. Equity 13.69%	U.S. REITs 2.52%	Commodity 19.53%	EM Equity 37.75%	Gold (1.56%)	U.S. Equity 31.49%	Gold 25.12%	U.S. REITs 43.06%	Gold (0.28%)	U.S. Equity 13.07%
U.S. REITs 8.69%	DM Equity 17.90%	DM Equity 23.29%	U.S. Bonds 5.97%	U.S. Equity 1.38%	U.S. Equity 11.96%	DM Equity 25.62%	U.S. Equity (4.38%)	U.S. REITs 25.84%	EM Equity 18.69%	Commodity 42.60%	U.S. Bonds (13.01%)	DM Equity 7.59%
U.S. Bonds 7.84%	U.S. REITs 17.77%	U.S. REITs 2.47%	Gold (1.44%)	U.S. Bonds 0.55%	EM Equity 11.60%	U.S. Equity 21.83%	U.S. REITs (4.57%)	DM Equity 22.66%	U.S. Equity 18.40%	U.S. Equity 28.71%	DM Equity (14.01%)	EM Equity 2.16
U.S. Equity 2.11%	U.S. Equity 16.00%	U.S. Bonds (2.02%)	EM Equity (1.82%)	DM Equity (0.39%)	U.S. REITs 8.60%	Gold 13.53%	Commodity (11.18%)	EM Equity 18.88%	DM Equity 8.28%	DM Equity 11.78%	U.S. Equity (18.11%)	Commodity 1.88%
Commodity (2.39%)	Gold 7.06%	EM Equity (2.27%)	DM Equity (4.48%)	Gold (10.41%)	Gold 8.14%	Commodity 6.16%	DM Equity (13.36%)	Gold 18.31%	U.S. Bonds 7.51%	U.S. Bonds (1.54%)	EM Equity (19.74%)	Gold 1.35%
DM Equity (11.73%)	U.S. Bonds 4.22%	Commodity (6.55%)	Commodity (26.43%)	EM Equity (14.60%)	U.S. Bonds 2.65%	U.S. REITs 5.07%	EM Equity (14.24%)	Commodity 12.94%	Commodity (7.53%)	EM Equity (2.22%)	U.S. REITs (24.51%)	U.S. Bonds (1.21%)
EM Equity (18.17%)	Commodity 4.16%	Gold (28.28%)	Bitcoin (57.74%)	Commodity (26.68%)	DM Equity 1.51%	U.S. Bonds 3.54%	Bitcoin (74.67%)	U.S. Bonds 8.72%	U.S. REITs (7.57%)	Gold (3.64%)	Bitcoin (63.90%)	U.S. REITs (1.95%)

Source: Bitwise Asset Management with data from Bloomberg. Data from December 31, 2010 to September 29, 2023.

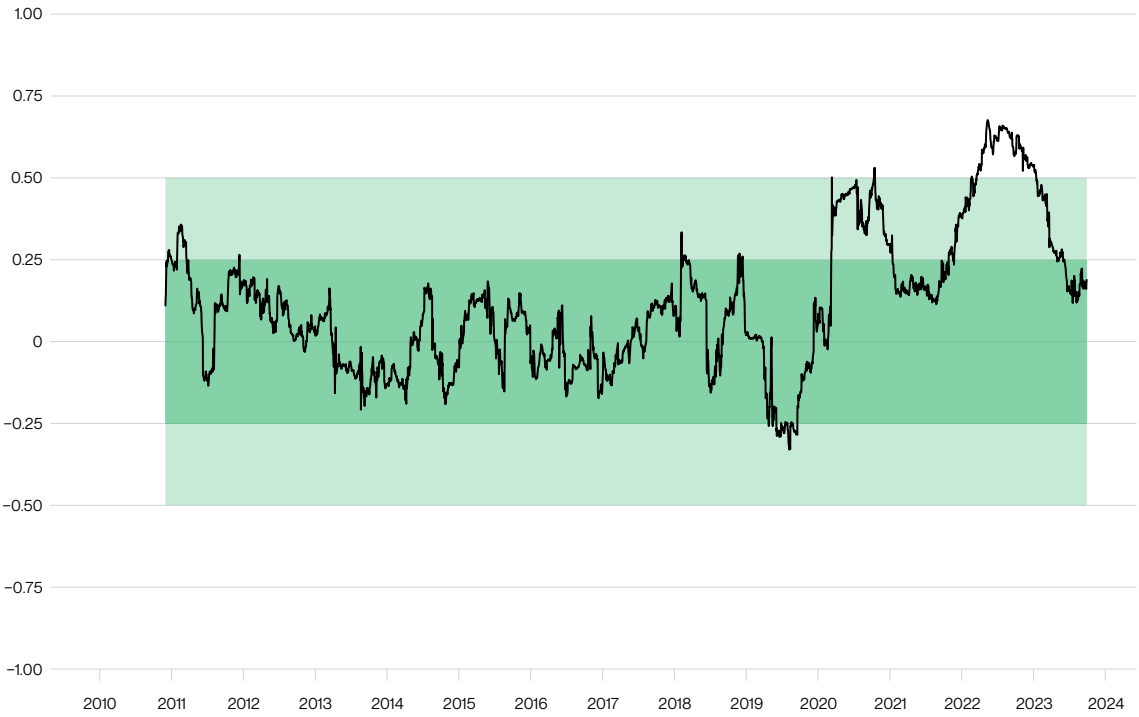
Note: Asset classes are represented by the following. Bitcoin: BTC spot price. Gold: Gold spot price. U.S. REITs: MSCI U.S. REIT Gross Total Return Index. U.S. Bonds: Bloomberg U.S. Aggregate Bond Index. U.S. Equity: S&P 500 Total Return Index. Commodity: Deutsche Bank DBIQ Optimum Yield Diversified Commodity Index Total Return. DM Equity: MSCI EAFE Gross Total Return USD Index. EM Equity: MSCI Emerging Markets Gross Total Return USD Index. All calculations are total return, including dividends for the stated period. For definitions of the indexes and assets listed here, see p.43.

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Correlations: Bitcoin vs. S&P 500

90-Day Moving Average



Source: Bitwise Asset Management with data from Bloomberg. Data from July 17, 2010 to September 29, 2023. For definitions of the indexes and assets listed here, see p.43.

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Correlations of Select Assets and Asset Classes

90-Day Moving Average

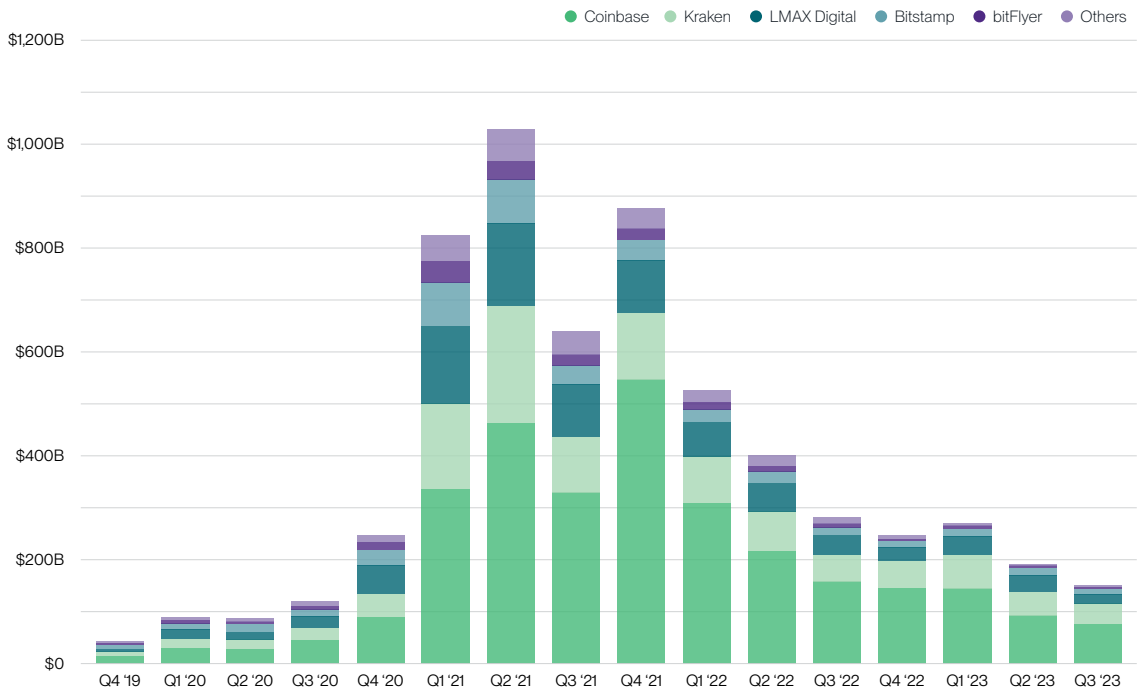
	Bitwise 10 Large Cap Crypto Index	Bitwise Crypto Innovators 30 Index	Bitcoin	U.S. Equities	Developed Market Equities	Emerging Market Equities	Commodities	U.S. REITs	U.S. Bonds	Gold
Bitwise 10 Large Cap Crypto Index	1.00									
Bitwise Crypto Innovators 30 Index	0.71	1.00								
Bitcoin	0.99	0.70	1.00							
U.S. Equities	0.21	0.41	0.19	1.00						
Developed Market Equities	0.16	0.25	0.13	0.52	1.00					
Emerging Market Equities	0.09	0.20	0.07	0.48	0.73	1.00				
Commodities	(0.03)	0.02	(0.05)	0.25	0.40	0.36	1.00			
U.S. REITs	0.15	0.28	0.14	0.52	0.46	0.49	0.20	1.00		
U.S. Bonds	0.19	0.34	0.19	0.22	0.32	0.16	0.10	0.20	1.00	
Gold	0.04	0.18	0.03	0.25	0.38	0.19	0.21	0.15	0.57	1.00

Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

Note: Asset classes are represented by the following. U.S. Equities: S&P 500 Total Return Index. DM Equities: MSCI EAFE Gross Total Return USD Index. EM Equities: MSCI Emerging Markets Gross Total Return USD Index. Commodities: Deutsche Bank DBIQ Optimum Yield Diversified Commodity Index Total Return. U.S. REITs: MSCI U.S. REIT Gross Total Return Index. U.S. Bonds: Bloomberg U.S. Aggregate Bond Index. Gold: Gold spot price. For definitions of the indexes and assets listed here, see p.43.

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Spot Trading Volume on Select Centralized Exchanges

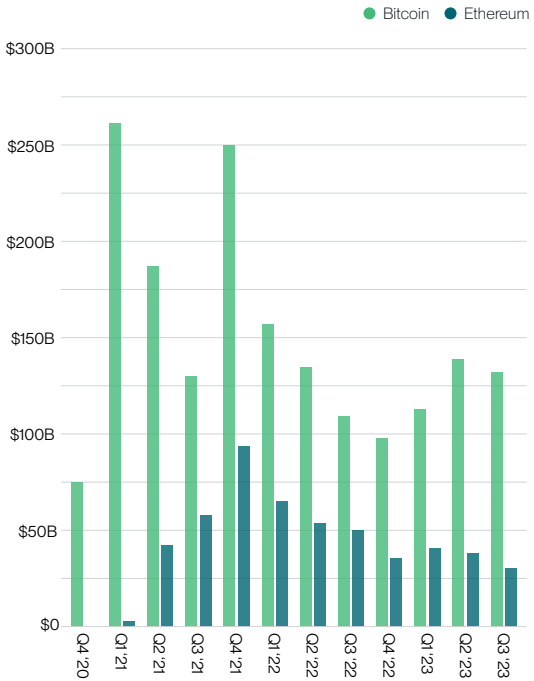


Source: Bitwise Asset Management with data from The Block. Data from Q4 2019 to Q3 2023.

Note: "Others" includes Gemini, Bittrex, and itBit. "Select Centralized Exchanges" refers to venues that meet the criteria for "Eligible Crypto Asset Trading Venues" per the Bitwise Crypto Asset Index Methodology. The methodology is available at https://bitwiseinvestments.com/indexes/methodology. "Spot trading" refers to trading in a financial market in which financial instruments are traded for immediate delivery.

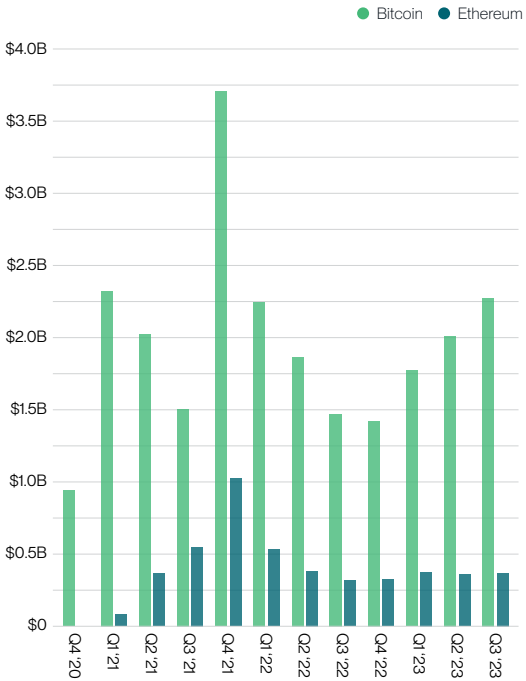
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Total Volume: CME Bitcoin and Ethereum Futures



Source: Bitwise Asset Management with data from The Block and CoinGlass. Data from Q4 2020 to Q3 2023.

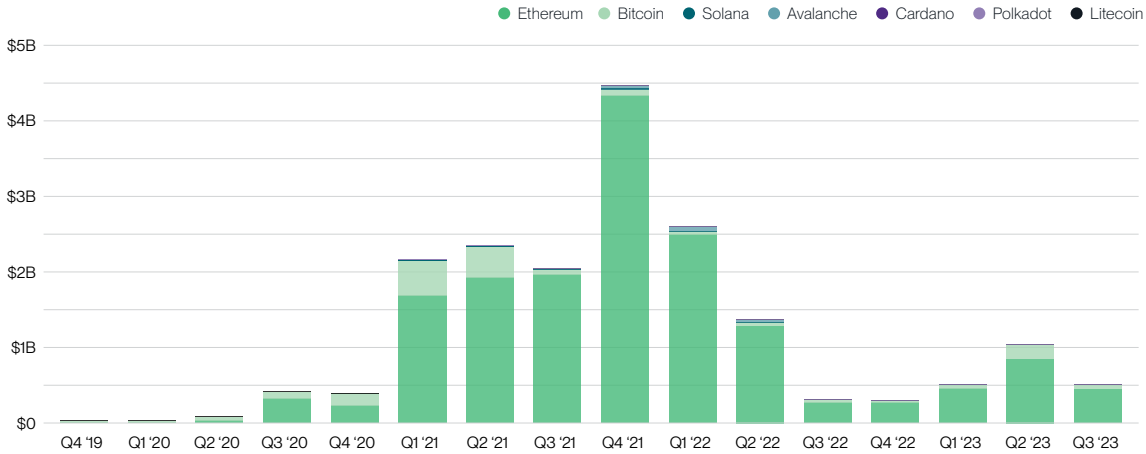
Average Open Interest: CME Bitcoin and Ethereum Futures



Source: Bitwise Asset Management with data from The Block and CoinGlass. Data from Q4 2020 to Q3 2023.

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Revenue: Leading Blockchains



— USD millions —

	2019	2020	2021	2022	2023 (YTD)
Ethereum (ETH)	34.67	593.76	9,918.09	4,315.38	1,751.20
Bitcoin (BTC)	155.35	324.35	1,018.50	142.05	295.98
Solana (SOL)	—	0.01	28.03	26.40	11.03
Avalanche (AVAX)	—	0.00	40.88	95.07	8.30
Cardano (ADA)	0.01	0.04	8.03	8.36	2.05
Polkadot (DOT)	—	0.02	2.13	0.73	0.31
Litecoin (LTC)	0.41	0.23	1.22	0.46	0.30

Source: Bitwise Asset Management with data from Token Terminal. Chart data from Q4 2019 to Q3 2023. Table data from January 1, 2019 to September 30, 2023.

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Fee Per Transaction: Bitcoin

30-Day Moving Average



Fee Per Transaction: Ethereum

30-Day Moving Average

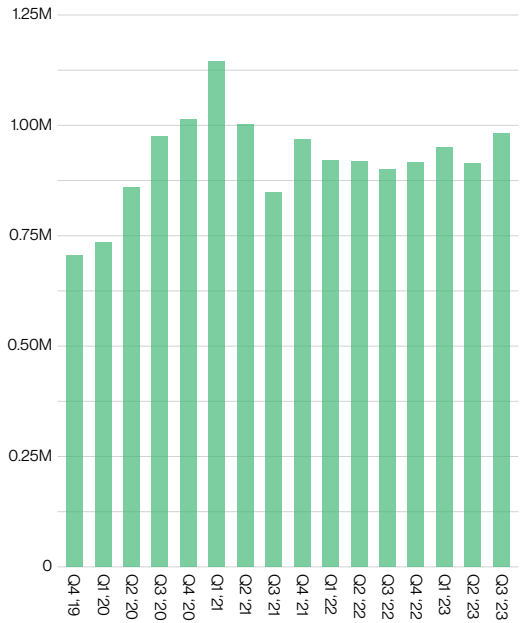


Source: Bitwise Asset Management with data from Coin Metrics. Data from Q4 2019 to Q3 2023.

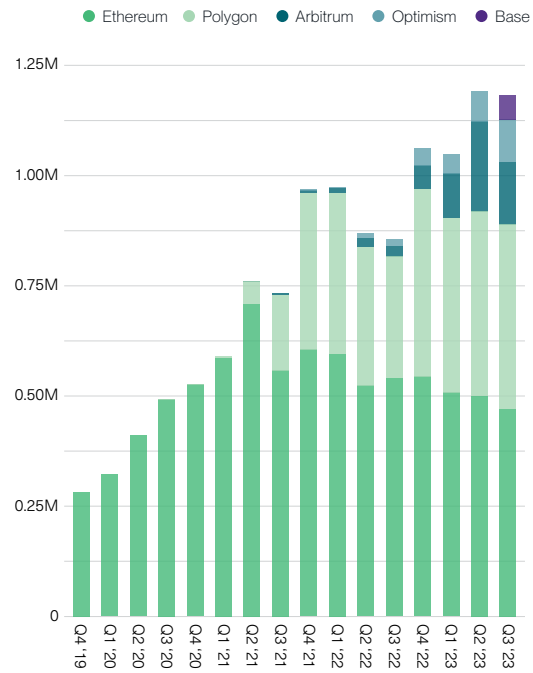
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Average Daily Active Users (On Chain)

Bitcoin



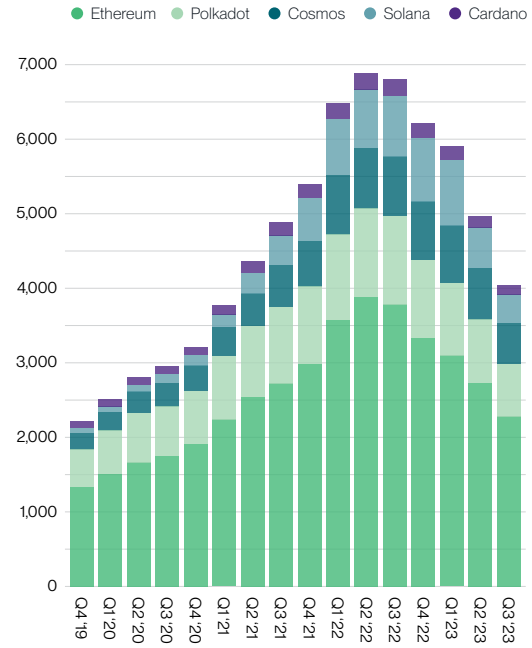
Ethereum Ecosystem



Source: Bitwise Asset Management with data from Coin Metrics and Token Terminal. Data from Q4 2019 to Q3 2023.
Note: "Ethereum Ecosystem" data includes Ethereum Layer 1 and select Ethereum Layer 2s.

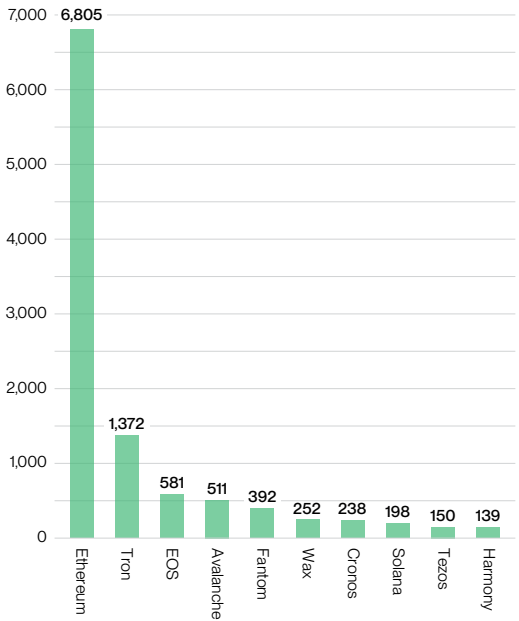
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Average Developer Count on Leading Blockchains



Source: Bitwise Asset Management with data from Artemis. Data from Q4 2019 to Q3 2023. Note: "Ethereum" data includes Ethereum Layer 1 and select Ethereum Layer 2s (Polygon, Optimism, and Arbitrum).

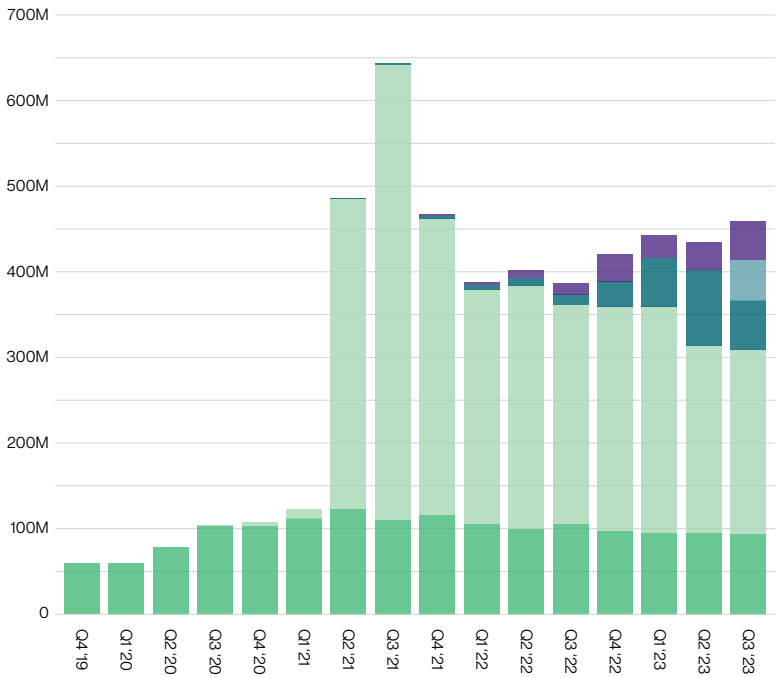
Number of Applications on Leading Blockchains



Source: Bitwise Asset Management with data from DappRadar. Data as of September 30, 2023. Note: "Ethereum" data includes Ethereum Layer 1 and select Ethereum Layer 2s (Polygon, Optimism, Arbitrum, and Base).

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Transaction Count: Ethereum Ecosystem



Asset	Median Fee per Transaction (Q3 2023)
Ethereum	$4.69
Arbitrum	$0.16
Optimism	$0.16
Base	$0.13
Polygon	$0.02

Source: Bitwise Asset Management with data from Token Terminal. "Transaction Count" data from Q4 2019 to Q3 2023. "Median Fee per Transaction" data from July 1, 2023 to September 30, 2023.

Note: "Ethereum Ecosystem" data includes Ethereum Layer 1 and select Ethereum Layer 2s.

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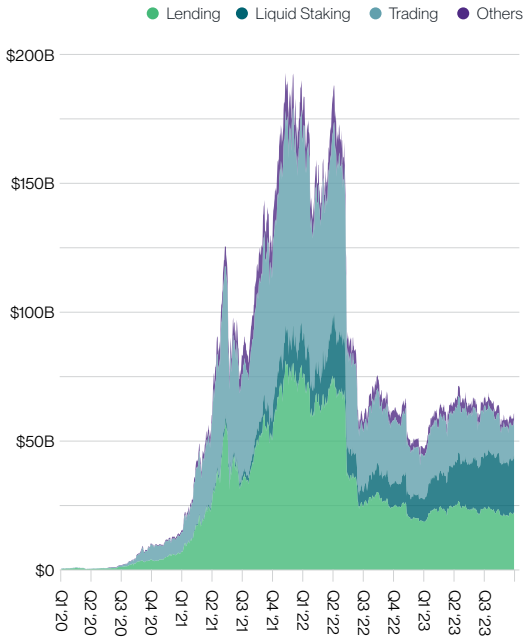
Top 10 Ethereum Applications by Revenue

Protocols	Description	TTM Revenue (USD millions)[1]
Uniswap (UNI)	Leading decentralized crypto exchange	532.80
GMX (GMX)	Leading crypto derivatives exchange	137.84
Aave (AAVE)	Largest lending and borrowing application in DeFi	91.67
Arbitrum (ARB)	A leading scaling solution for Ethereum	66.62
MakerDAO (MKR)	Lending app behind largest crypto-backed stablecoin	62.84
Curve (CRV)	Leading decentralized exchange for stablecoins	62.56
SushiSwap (SUSHI)	Popular platform for launching and trading digital assets	55.70
Polygon (MATIC)	Largest network of Ethereum scaling solutions	34.46
Synthetix (SNX)	Popular platform for creating synthetic digital assets	28.30
KyberSwap (KNC)	Popular decentralized exchange for trading digital assets	7.17

Source: Bitwise Asset Management with data from Token Terminal. Data from October 1, 2022 to September 30, 2023.

(1) Revenue is made up of the total fees paid by users. TTM stands for "trailing twelve months," i.e., each protocol's revenue over the last four quarters.

Total Value Locked in DeFi by Sector

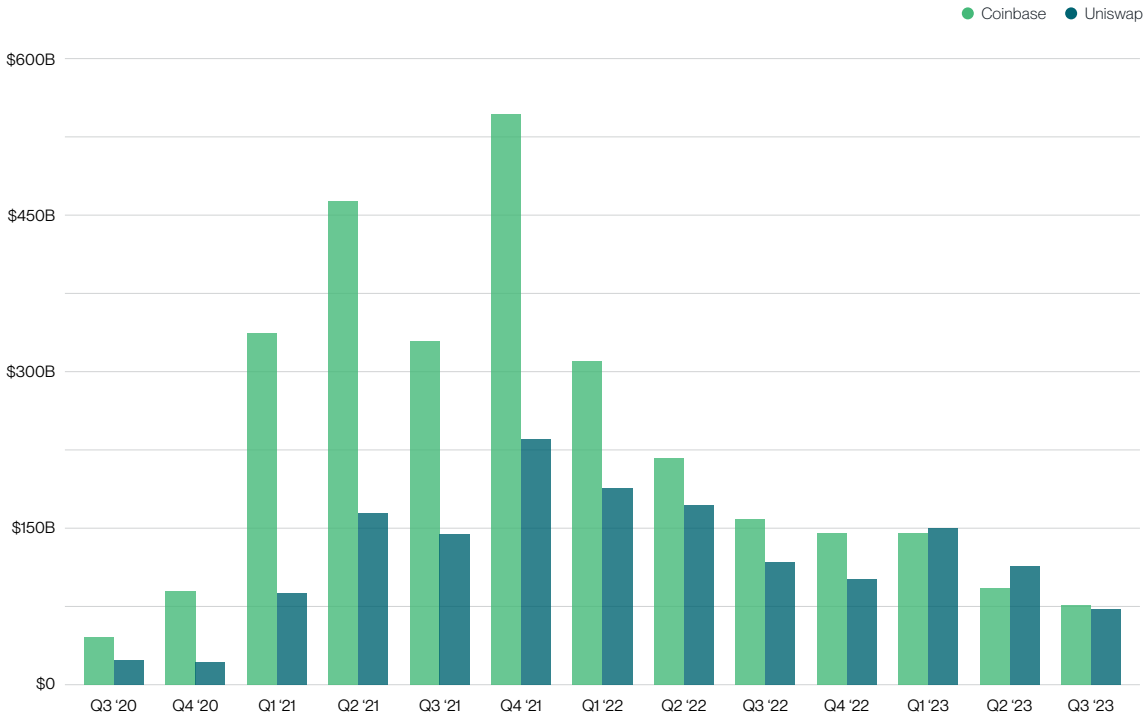


Source: Bitwise Asset Management with data from The Block and DeFi Llama. Data from Q1 2020 to Q3 2023.

Note: Total Value Locked (TVL) is a measure of the total amount of capital deployed in DeFi applications.

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Trading Volume: Coinbase vs. Uniswap



Source: Bitwise Asset Management with data from The Block, The Graph, and CoinGecko. Data from Q3 2020 to Q3 2023.

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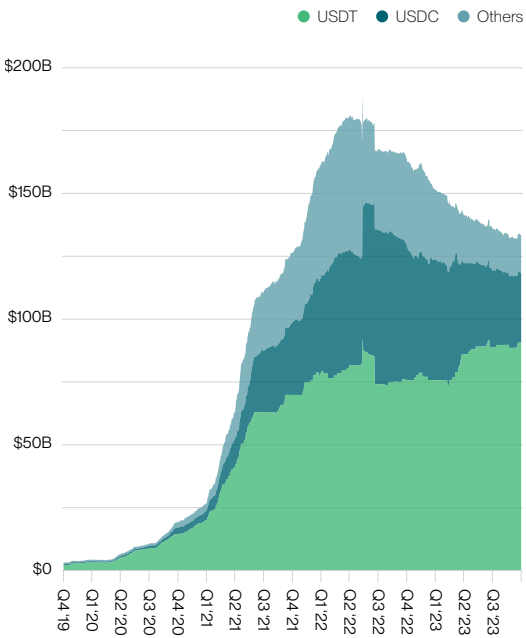
Total Deposits and Outstanding Loans in Leading DeFi Lending Protocols



Source: Bitwise Asset Management with data from The Block. Data from Q4 2019 to Q3 2023.
Note: Data includes only the leading Ethereum-based DeFi lending protocols: Aave, MakerDAO, and Compound.

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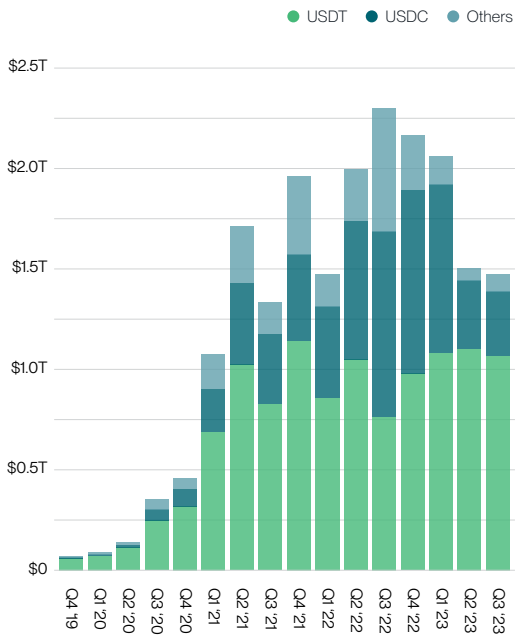
Stablecoin Market Capitalization



Source: Bitwise Asset Management with data from The Block, Coin Metrics, and CoinGecko. Data from Q4 2019 to Q3 2023.

Note: "Others" includes BUSD, crvUSD, DAI, FDUSD, FEI, FRAX, GHO, GUSD, HUSD, LUSD, PYUSD, TUSD, and USDP.

Stablecoin Transfer Volume

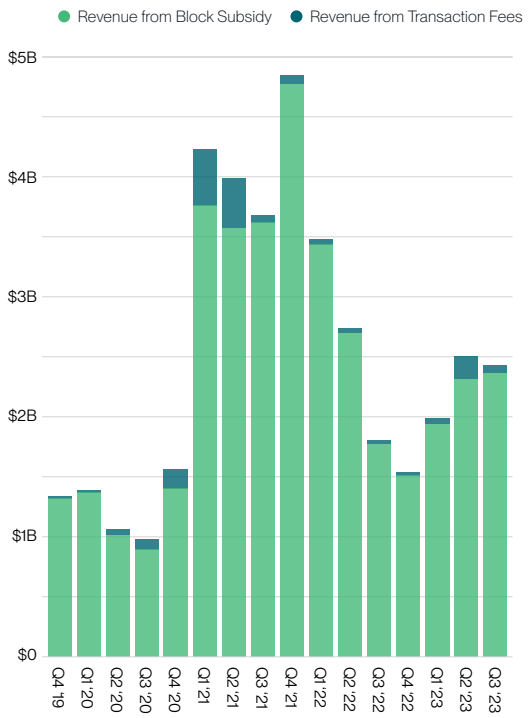


Source: Bitwise Asset Management with data from Coin Metrics. Data from Q4 2019 to Q3 2023.

Note: "Others" includes BUSD, DAI, GUSD, HUSD, PAXG, TUSD, USDK, and USDP.

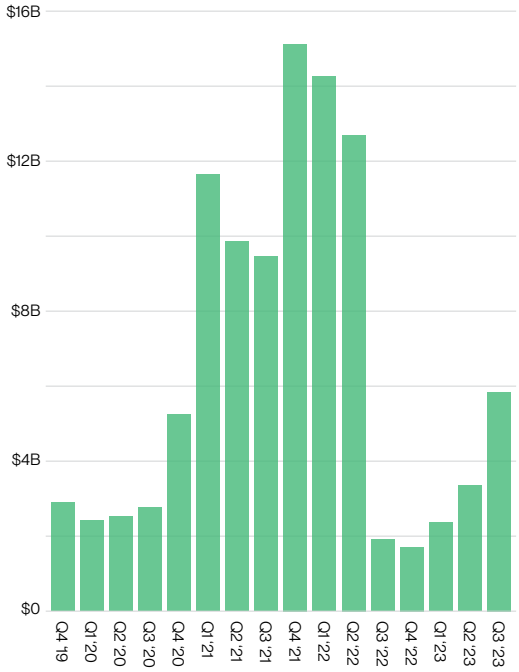
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Bitcoin Miner Revenue by Type



Source: Bitwise Asset Management with data from The Block. Data from Q4 2019 to Q3 2023.

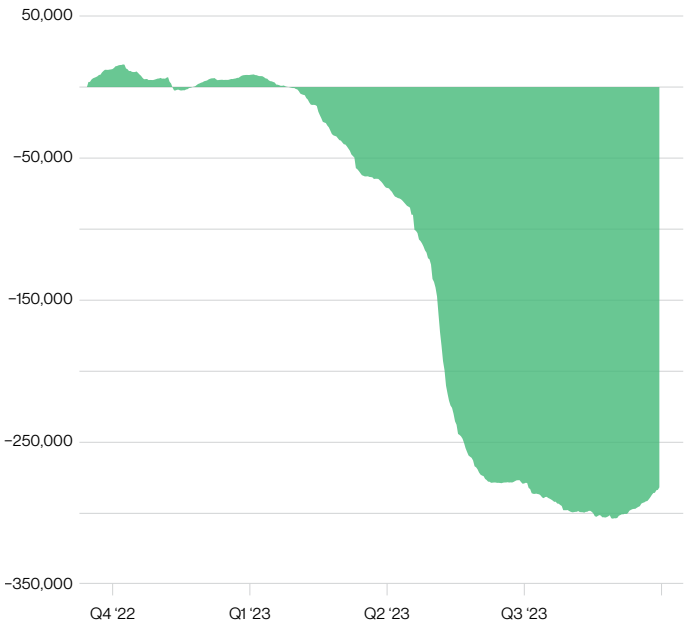
Value of Bitcoin Flows From Miners to Exchanges



Source: Bitwise Asset Management with data from Coin Metrics. Data from Q4 2019 to Q3 2023.

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Change in Ethereum Supply Since Transition to Proof-of-Stake ("The Merge")



Source: Bitwise Asset Management with data from Ultra Sound Money. Data from September 15, 2022 to September 30, 2023.

Note: As of September 29, 2023, the total supply of Ethereum was 120,235,706 ETH.

Amount of Ethereum Burned

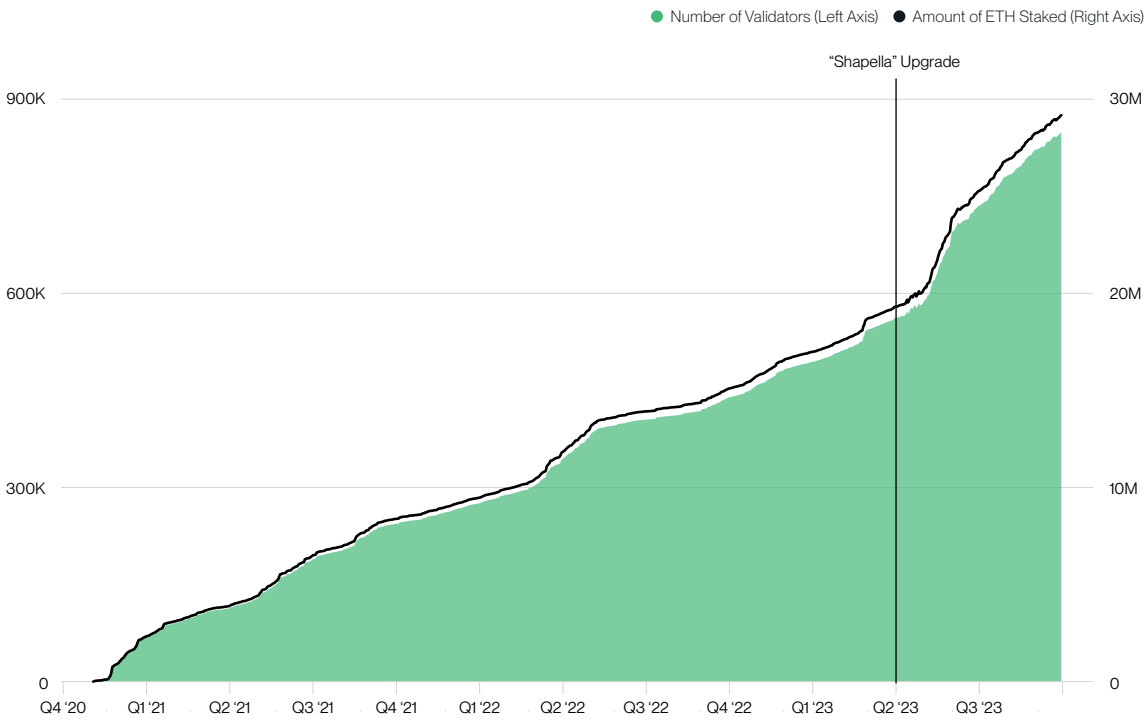
Time Frame	ETH Burned
Last 30 days	**44,301**
Since "The Merge"	**1,000,310**
Since EIP-1559	**3,632,619**

Source: Bitwise Asset Management with data from Ultra Sound Money. Data as of September 30, 2023

Note: EIP-1559 was implemented on August 5, 2021. The Merge took place on September 15, 2022.

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Amount of Ethereum Staked and Number of Ethereum Validators



Source: Bitwise Asset Management with data from Dune Analytics. Data from November 3, 2020 to September 30, 2023.

Note: Staking involves making a financial commitment to a blockchain in its native asset to secure the network, typically in exchange for yield. Staking typically requires the owner to lock up the staked asset for a period of time. The "Shapella" upgrade in April 2023 allowed staked ETH to be withdrawn.

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Top Crypto Equities: Fundamental Metrics

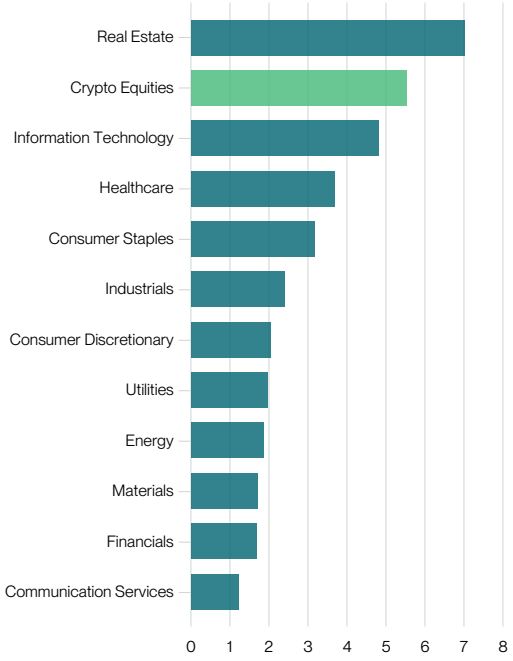
			Revenue (USD millions)				*Net Income (USD millions)*			
Asset	Market Cap (USD millions)	Price/Sales Ratio	Q3 2022	Q4 2022	Q1 2023	Q2 2023	Q3 2022	Q4 2022	Q1 2023	Q2 2023
Coinbase (COIN)	$17,800	6.38	$590	$629	$773	$708	($545)	($557)	($79)	($97)
MicroStrategy (MSTR)	$4,620	7.79	$125	$133	$122	$120	($27)	($250)	$461	$22
Riot Platforms (RIOT)	$1,730	5.90	$46	$60	$73	$77	($37)	($142)	($56)	($28)
Marathon Digital (MARA)	$1,480	7.44	$13	$28	$51	$82	($75)	($407)	($7)	($19)
Galaxy Digital (GLXY CN)	$1,210	3.10	$33	$157	$147	$109	($68)	($689)	$134	($46)
Bitdeer Technologies (BTDR)	$1,070	—	—	—	$73	$94	—	—	($9)	($40)
Applied Digital (APLD)	$663	10.80	$7	$12	$14	$22	($5)	($27)	($7)	($6)
Hut 8 Mining (HUT CN)	$437	5.99	$24	$16	$14	—	($18)	($138)	$80	—
Canaan (CAN)	$347	14.13	$138	$57	$55	$74	$9	($64)	($84)	($111)
Bitfarms (BITF CN)	$230	1.87	$33	$27	$30	$35	($85)	($17)	($2)	($25)

Source: Bitwise Asset Management with data from Bloomberg and company filings. Market Cap and Price/Sales Ratio data as of September 29, 2023. Revenue and Net Income data from July 1, 2022 to June 30, 2023.

Note: "Top Crypto Equities" refers to select "pure-play," or primarily crypto-focused, constituents of the Bitwise Crypto Innovators 30 Index. Publicly traded equities report data with a significant lag, so the most recently available data is from Q2 2023. Applied Digital reports quarterly results on a fiscal year basis instead of a calendar year basis. Due to this, its quarterly reports have a one-month difference with calendar year quarterly reports (i.e., Q2 2023 is as of May 2023 instead of June 2023). Data for Hut 8 Mining is as of March 31, 2023 as the company is in the middle of a merger with U.S. Bitcoin. Price/Sales ratio uses trailing 12-month sales.

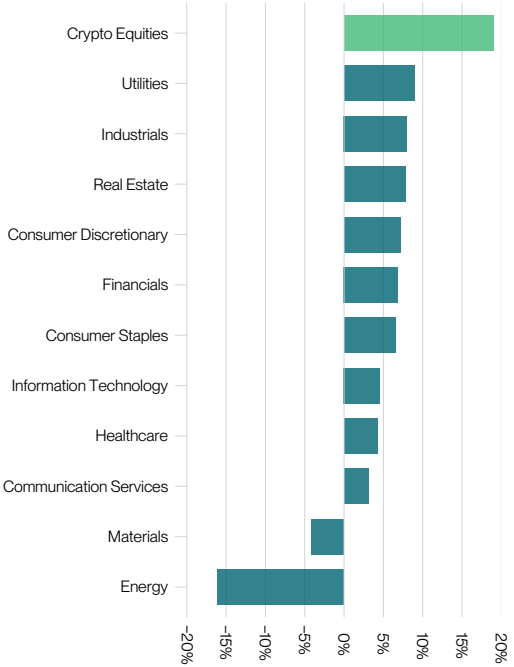
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Crypto Equities vs. Other Industries: Price-to-Sales Ratio



Source: Bitwise Asset Management with data from Bloomberg. Price data as of September 29, 2023. Sales data is trailing 12-month sales as of June 30, 2023.

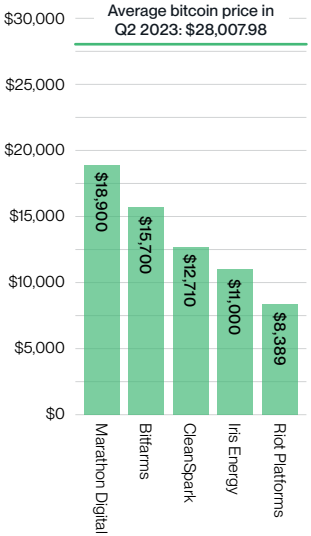
Crypto Equities vs. Other Industries: 2023 Estimated Full-Year Revenue Growth



Source: Bitwise Asset Management with data from Bloomberg. Data as of September 29, 2023.

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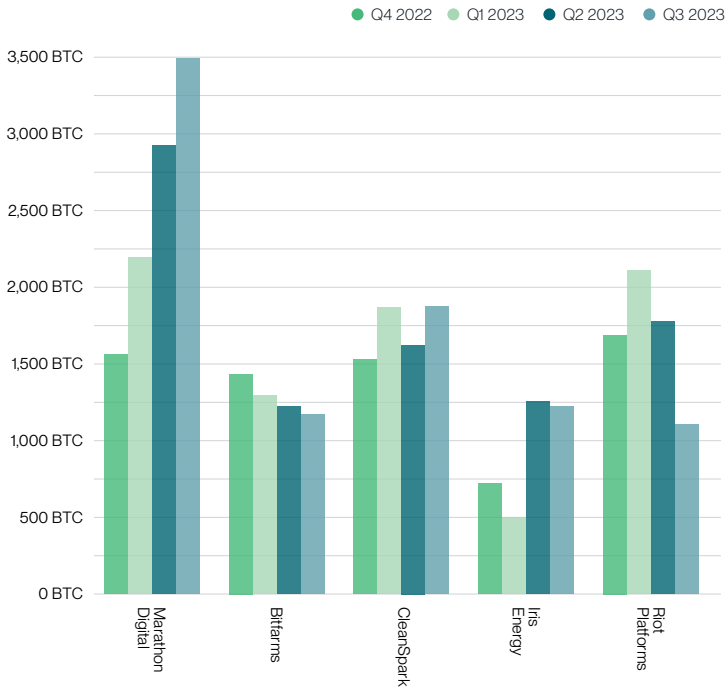
Average Mining Cost Per Bitcoin: Top Five Miners



Source: Bitwise Asset Management with data from company filings. Data from April 1 to June 30, 2023.

Note: "Top five miners" refers to the top five miners by bitcoin production in the most recent quarter. Iris Energy's $11,000 mining cost per bitcoin is the full year figure instead of the quarter due to the company's reporting schedule.

Bitcoin Production by Top Five Miners



Source: Bitwise Asset Management with data from company filings. Data from October 1, 2022 to September 30, 2023.

Note: "Top five miners" refers to the top five miners by bitcoin production in the most recent quarter.

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Crypto Mining Companies: Core Fundamental Metrics

Company	Net Debt/Equity Ratio	Current Ratio	Operating Cash Flow Ratio
Bit Digital (BTBT)	(0.57)	10.31	(1.44)
Bitfarms (BITF CN)	(0.11)	1.93	0.04
Cipher Mining (CIFR)	(0.04)	1.09	0.36
CleanSpark (CLSK)	(0.03)	1.41	0.04
HIVE Blockchain (HIVE CN)	(0.28)	3.12	0.67
Hut 8 Mining (HUT CN)	(0.73)	9.85	(0.24)
Iris Energy (IREN)	(0.22)	3.72	—
Marathon Digital (MARA)	0.65	14.91	(4.02)
Riot Platforms (RIOT)	(0.35)	6.49	(0.51)
TeraWulf (WULF)	0.82	0.18	(0.00)
Average	**(0.09)**	**5.30**	**(0.57)**
Median	**(0.17)**	**3.42**	**(0.00)**

Source: Bitwise Asset Management with data from company filings. Data as of June 30, 2023.

Note: Data for Hut 8 Mining is as of March 31, 2023 as the company is in the middle of a merger with U.S. Bitcoin. Net Debt/Equity ratio includes the value of digital assets as part of cash and cash equivalents. Iris Energy did not report Cash Flow from Operations for Q2 2023, only full-year results.

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Coinbase: Select Annual and Year-End Metrics

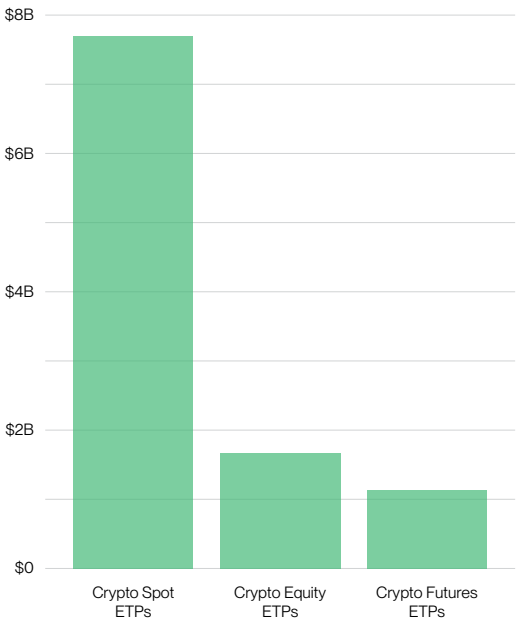
	2019	2020	2021	2022	2023 (Est.)
Transaction Revenue (USD millions)	$463	$1,096	$6,837	$2,356	$1,404[1]
Net Revenue (USD millions)	$483	$1,141	$7,355	$3,149	$2,798[1]
Net Income (USD millions)	($30)	$322	$3,624	($2,625)	($352)[1]
Trading Volume (USD billions)	$80	$193	$1,671	$830	$474[1]
Assets on Platform (USD billions)	$17	$90	$278	$80	$128[2]
Monthly Transacting Users (millions)	1.0	2.8	11.2	8.3	7.9[2]

Source: Bitwise Asset Management with data from company filings. Data from January 1, 2019 to June 30, 2023.

Notes: (1) Estimated 2023 values for Transaction Revenue, Net Revenue, Net Income, and Trading Volume are annualized based on data from January 1 to June 30, 2023. (2) Assets on Platform and Monthly Transacting Users are as of June 30, 2023.

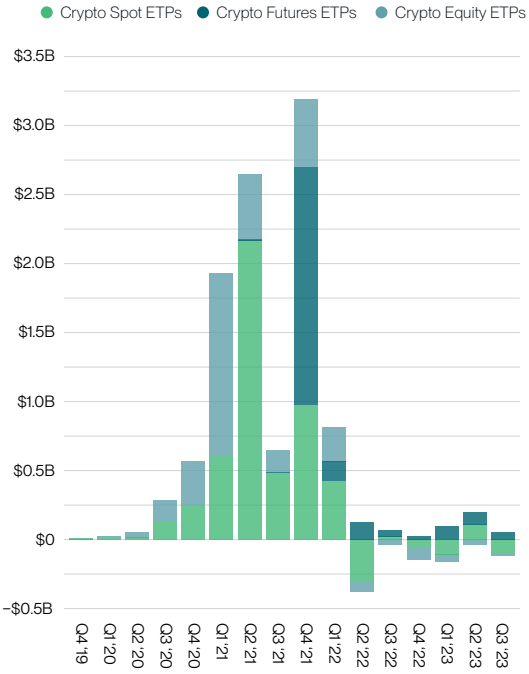
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Crypto Exchange-Traded Products (Global): AUM



Source: Bitwise Asset Management with data from Bloomberg and fund documents. Data as of September 29, 2023.

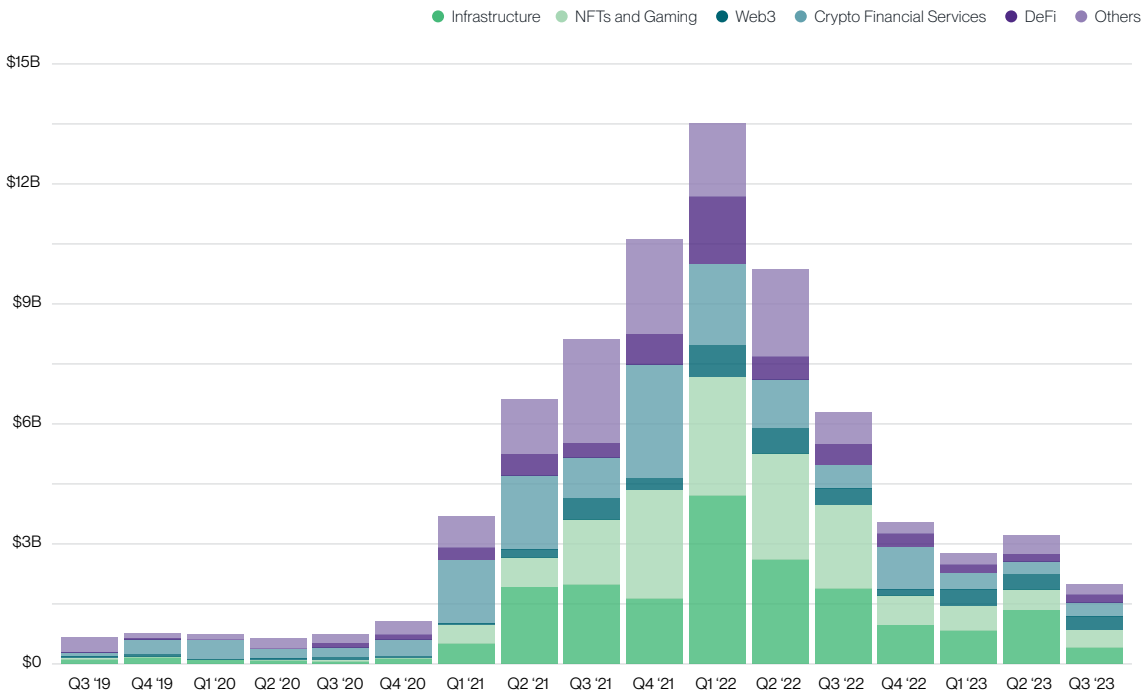
Crypto Exchange-Traded Products (Global): Net Flows



Source: Bitwise Asset Management with data from Bloomberg. Data from Q4 2019 to Q3 2023.

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Crypto Venture Funding



Source: Bitwise Asset Management with data from The Block. Data from Q3 2019 to Q3 2023.

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Crypto Adoption by Traditional Financial Institutions

Financial Institution	Crypto Activity	Investable Client Assets (USD billions)
BlackRock	Filed for spot Bitcoin ETF in U.S.; manages USDC reserves fund	**9,425.21**
Charles Schwab	Launched crypto exchange in partnership with Fidelity and Citadel Securities	**8,015.80**
Fidelity	Filed for spot Bitcoin ETF in U.S. ; supports Bitcoin and Ethereum trading and custody	**4,470.90**
JPMorgan Chase	Supports tokenized USD and EUR transfers via JPM Coin	**3,200.00**
Goldman Sachs	Supports OTC crypto trading; structured $100 million bond on Ethereum	**2,714.00**
UBS	Exploring tokenized bonds and money markets on Ethereum	**2,568.00**
BNY Mellon	Supports holding and transferring Bitcoin and Ethereum	**1,906.00**
Invesco	Filed for spot Bitcoin ETF in U.S.; offers spot Bitcoin ETF in Europe	**1,538.20**
Bank of America	Supports Bitcoin futures trading	**1,531.00**
Deutsche Bank	Developing digital asset custody and tokenization services	**1,521.31**
Franklin Templeton	Operates a tokenized money market fund; filed for spot Bitcoin ETF in U.S.	**1,431.50**
Morgan Stanley	Provides access to multiple Bitcoin funds	**1,412.00**
Total		**39,733.92**

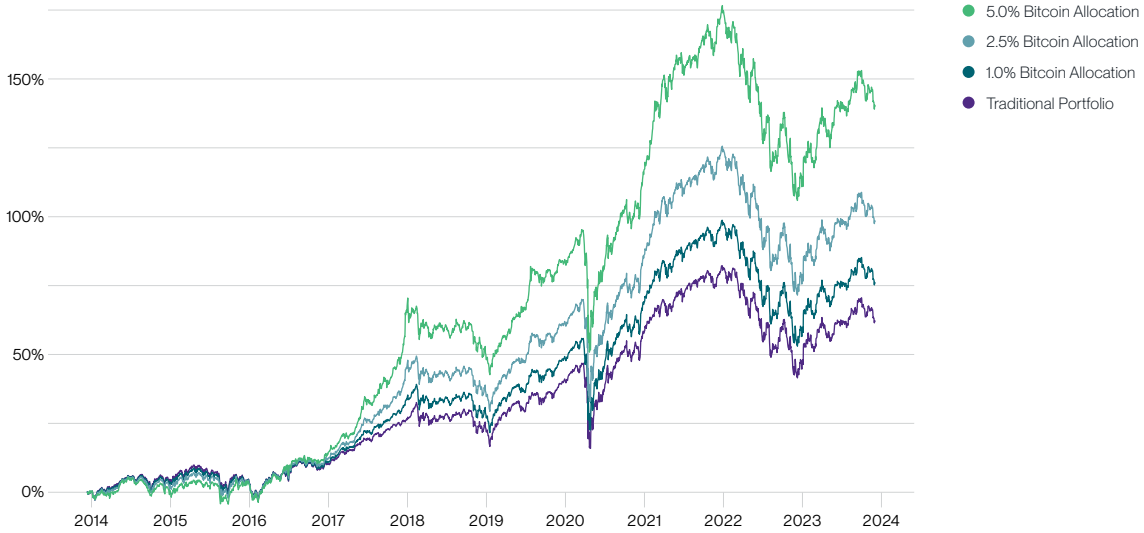
Source: Bitwise Asset Management with data from company filings and presentations. Data as of June 30, 2023.

Note: This table was inspired by and builds on a tweet by Meltem Demirors, the Chief Strategy Officer of CoinShares. "Investable Client Assets" refers to the assets under management of the institution's wealth and asset management divisions only, if that information is broken out in the institution's reporting.

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Bitcoin's Historical Impact on a Traditional 60/40 Portfolio: Performance

Cumulative Returns: Traditional Portfolio With and Without Quarterly Rebalanced Bitcoin Allocations



Source: Bitwise Asset Management with data from IEX Cloud. Data from January 1, 2014 to September 29, 2023.

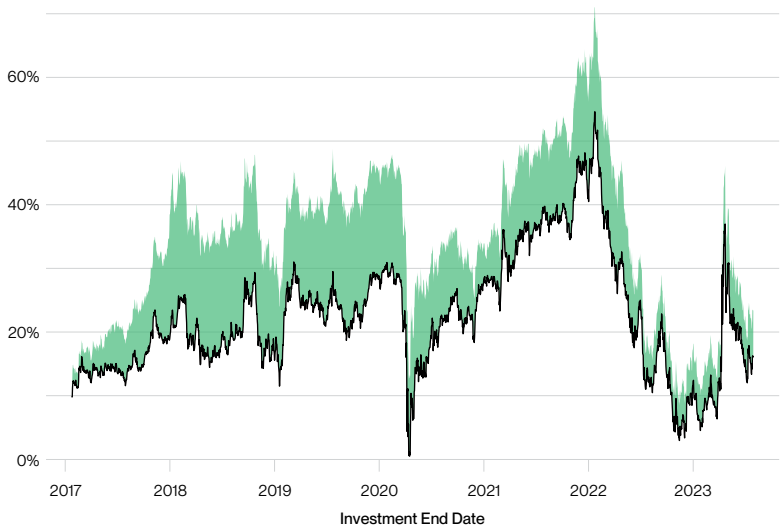
Note: Traditional Portfolio consists of 60% equities (represented by the Vanguard Total Stock ETF, VT) and 40% bonds (represented by the Vanguard Total Bond ETF, BND). Not considering taxes nor transaction costs. Bitcoin (BTC) is analyzed here because its longer price history allows for more comprehensive historical analysis than other crypto assets. Performance of individual crypto assets may differ significantly from the performance of bitcoin. For more details, please refer to our white paper "Bitcoin's Role in a Traditional Portfolio" published in August 2023 and available at https://bitwiseinvestments.com/crypto-market-insights/bitcoins-role-in-a-traditional-portfolio.

Past performance does not predict or guarantee future results. Nothing contained herein is intended to predict the performance of any investment. There can be no assurance that actual outcomes will match the assumptions or that actual returns will match any expected returns. Historical performance of sample portfolios has been generated and maximized with the benefit of hindsight. The returns do not represent the returns of an actual account and do not include the fees and expenses associated with buying, selling and holding funds or crypto assets. Performance information is provided for informational purposes only. Please see important disclosure information in this document.

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Bitcoin's Historical Impact on a Traditional 60/40 Portfolio: Win Rate

Contribution of a 2.5% Bitcoin Allocation to a 60/40 Portfolio: 3-Year Rolling Cumulative Return



Maximum contribution	22.31pp
Median contribution	12.08pp
Minimum contribution	1.84pp
Win Rate	100%
Loss Rate	0%

Source: Bitwise Asset Management with data from IEX Cloud. Data from January 1, 2017 to September 29, 2023.

Note: Traditional Portfolio consists of 60% equities (represented by the Vanguard Total Stock ETF, VT) and 40% bonds (represented by the Vanguard Total Bond ETF, BND). Not considering taxes nor transaction costs. Bitcoin (BTC) is analyzed here because its longer price history allows for more comprehensive historical analysis than other crypto assets. Performance of individual crypto assets may differ significantly from the performance of bitcoin. For more details, please refer to our white paper "Bitcoin's Role in a Traditional Portfolio" published in August 2023 and available at https://bitwiseinvestments.com/crypto-market-insights/bitcoins-role-in-a-traditional-portfolio.

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Status of Spot Bitcoin ETF Applications in the U.S.

Issuer (Ticker)	Company	First Deadline	Second Deadline	Third Deadline	Final Deadline
ARK 21Shares Bitcoin ETF (ARKB)	21Shares and ARK	6/29/23	8/13/23	11/11/23	1/10/24
Bitwise Bitcoin ETP Trust	Bitwise	9/1/23	10/16/23	1/14/24	3/14/24
iShares Bitcoin Trust	BlackRock	9/2/23	10/17/23	1/15/24	3/15/24
VanEck Bitcoin Trust	VanEck	9/2/23	10/17/23	1/15/24	3/15/24
WisdomTree Bitcoin Trust (BTCW)	WisdomTree	9/2/23	10/17/23	1/15/24	3/15/24
Invesco Galaxy Bitcoin ETF	Invesco and Galaxy Digital	9/2/23	10/17/23	1/15/24	3/15/24
Wise Origin Bitcoin Trust	Fidelity	9/2/23	10/17/23	1/15/24	3/15/24
Valkyrie Bitcoin Fund (BRRR)	Valkyrie	9/4/23	10/19/23	1/17/24	3/19/24
Global X Bitcoin Trust	Global X	10/7/23	11/21/23	2/19/24	4/19/24
Hashdex Bitcoin ETF *Strategy Change* (DEFI)	Hashdex	(est.) 11/17/23	(est.) 1/1/24	(est.) 3/31/24	(est.) 5/30/24
Franklin Bitcoin ETF	Franklin Templeton	(est.) 11/17/23	(est.) 1/1/24	(est.) 3/31/24	(est.) 5/30/24
Grayscale Bitcoin Trust *Conversion* (GBTC)	Grayscale	12/23/21	2/6/22	5/7/22	TBD

Scan to receive real-time news updates on filings to launch Bitcoin & Ethereum ETFs.

Source: Bitwise Asset Management with data from Bloomberg Intelligence. Data as of September 29, 2023.
Notes: Dates are estimates and/or deadlines. Actual decisions may come earlier.

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Recent U.S. Regulatory Developments

July and October 2023	**Ruling: SEC v. Ripple Labs**	In July, a judge in the Southern District of New York issued a summary judgment order in the SEC's lawsuit against Ripple Labs (filed in December 2020) that centered around whether a crypto asset developed by the creators of Ripple Labs, XRP, is a security. The court ruled that XRP is not a security because the sale of the digital token on public exchanges complied with federal securities laws. In October, the Court dismissed the SEC's motion to appeal the ruling.
July 2023	**Financial Innovation and Technology for the 21st Century Act Introduced**	The Financial Innovation and Technology for the 21st Century Act aims to define the regulatory status of crypto assets and the respective oversight roles of the CFTC and SEC. The Act passed both the House Financial Services and Agriculture Committees and moved to the House of Representatives for debate.
July 2023	**Responsible Financial Innovation Act Introduced**	The Responsible Financial Innovation Act is a comprehensive crypto regulatory bill that would, among other things, create a definition for decentralized finance and address issues ranging from anti-money laundering provisions to custody rules.
August 2023	**Clarity for Payment Stablecoins Act Introduced**	The Clarity for Payment Stablecoins Act was approved by the House Financial Services Committee and moved to the House of Representatives for debate. The bill aims to create a regulatory framework for the issuance and oversight of payment stablecoins in the U.S.
September 2023	**Ruling: Grayscale Investments v. SEC**	A three-judge panel ruled unanimously in favor of Grayscale in its lawsuit with the SEC over attempts to convert the Grayscale Bitcoin Trust (GBTC) into an ETF. The court found that the Grayscale ETF was treated differently than similar products, such as Bitcoin futures ETFs, and ruled that the SEC must review its rejection of Grayscale's application.
September 2023	**Digital Dollar Pilot Prevention Act Introduced**	The Digital Dollar Pilot Prevention Act would prevent the Federal Reserve from testing digital currencies without getting permission from Congress.
September 2023	**Amendment to the Federal Reserve Act Introduced**	An amendment to the Federal Reserve Act was introduced that would stop stop Federal Reserve banks from directly offering certain services to individuals. The bill would also prohibit the use of digital currencies for managing the money supply and other activities.

Source: Bitwise Asset Management.

Definitions

1) The **Bitwise 10 Large Cap Crypto Index** (XBITW) captures the 10 largest eligible crypto assets by free-float-adjusted market capitalization.

2) The **Bitwise Crypto Innovators 30 Total Return Index** (XBITQG) is an equity index that provides focused exposure to companies that are building the future of the crypto asset-enabled decentralized economy.

3) The **Bitwise Decentralized Finance Crypto Index** (XBITDEFI) tracks the value of crypto assets in the rapidly emerging Decentralized Finance space.

4) The **S&P 500® Total Return Index** (SPXT) tracks the performance of 500 large-cap publicly traded companies in the U.S.

5) The **MSCI U.S. REIT Gross Total Return Index** (RMSG) is a free float-adjusted market capitalization-weighted index that is comprised of equity Real Estate Investment Trusts (REITs).

6) The **Bloomberg U.S. Aggregate Bond Index** (LBUSTRUU) tracks U.S. dollar-denominated, investment-grade debt.

7) The **MSCI EAFE Gross Total Return USD Index** (M2EA) is designed to represent the performance of large- and mid-cap securities across 21 developed markets, including countries in Europe, Australasia and the Far East, excluding the U.S. and Canada.

8) The **MSCI Emerging Markets Gross Total Return USD Index** (M2EF) captures large- and mid-cap representation across 24 Emerging Markets (EM) countries.

9) The **Deutsche Bank DBIQ Optimum Yield Diversified Commodity Index Total Return** (DBLCDBCT) is based on 14 commodities drawn from the energy, precious metals, industrial metals and agriculture sectors.

10) "Gold" is represented by the spot gold price with data from Bloomberg.

11) Bitcoin and other crypto assets are represented by the respective spot price.

Risks and Important Information

The opinions expressed herein are intended to provide insight or education and are not intended as individual investment advice. Bitwise does not represent that this information is accurate and complete and should not be relied upon as such.

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